

NDX: Neutron Dose Rate Meters with Extended Capabilities

Pavel V. Degtiarenko

Abstract— The novel design of the neutron moderator for the NDX dose rate meter implements a layer of a Beryllium-loaded medium Z material which allows the device to detect higher energy neutron flux and convert it to dose. Using Beryllium helps to fill the known dip in the response functions around 5-50 MeV. Optimized design parameter set, obtained using FLUKA Monte Carlo simulations, features the NDX moderator made of polyethylene embedding a layer of Be-Cu alloy inside, with the dose rate response constant within $\pm 10\%$ in the neutron energy range from about 100 keV to 500 MeV, and $\pm 25\%$ essentially in the full energy range above thermal. A new sensor technique implemented in the NDX is based on using the Ionization Chamber (IC) filled with ^3He gas to capture the moderated neutron fluence, and includes a symmetric IC filled with regular ^4He to evaluate the contributions due to the non-neutron ionizing radiation. Advantages of the method include large accessible dynamic range, relative insensitivity of the IC probes to photon radiation fields, and the ability to subtract their contribution. Two prototype NDX detectors were built, calibrated, and operated in various conditions at the CEBAF 12 GeV electron accelerator at JLab. The first results showed reliable neutron dose rate measurements in the presence of overwhelming photon radiation fields, and at practical high dose rate levels of up to approximately 10 Sv/h. Detector design, operation, first results, and further development plans are presented.

Index Terms— detectors for mixed radiation fields, neutron detectors, neutron dose rate meters, radiation hard detectors, wide energy range and high dynamic range neutron meters.

I. INTRODUCTION

PARTICLE accelerators, irradiation facilities, and nuclear power plants would benefit from using neutron dose rate meters featuring improved energy response covering full range of expected neutron energies, improved dynamic range of measured dose rates while providing the real-time data, and improved ability to operate in harsh radiation environments, including heavy photon backgrounds and pulsed beams. Novel solutions implemented in the NDX dose rate meter design may fully or partially satisfy these extended requirements.

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P. V. Degtiarenko is with the Radiation Control Department, Thomas Jefferson National Accelerator Facility (Jefferson Lab), Newport News, VA 23606, USA (e-mail: pavel@jlab.org).

II. NDX NEUTRON DOSE RATE METER DESIGN

Two novel features, now patented in the U.S. [1], are implemented in the design of the NDX neutron detector, extending the capabilities of neutron dose rate meters available on the market presently. The first is the use of a layer of the Beryllium-loaded medium Z material, such as Be-Cu alloy, inside of a typical neutron moderator made of polyethylene. Using heavier Z material layers in neutron moderators, as compared to the classical neutron moderator design [2], is known [3-5] to increase the detector sensitivity to higher neutron energies. Beryllium helps to increase sensitivity and further improve the detector's energy response in the energy range around and just above 10 MeV.

The second novel feature is the use of symmetric Ionization Chambers (IC) filled with different isotopes of Helium gas at the same pressure, as the sensors in the NDX detector. Captures of thermalized neutrons in ^3He produce significant gas ionization that can be measured by the electrometer-type high sensitivity readout electronics. ^4He is practically insensitive to thermalized neutrons. Both ICs are equally sensitive to photons, other ionizing radiation, and higher energy neutron recoils. The difference between the readout currents is therefore proportional to the thermalized neutron fluence in the core of the moderator.

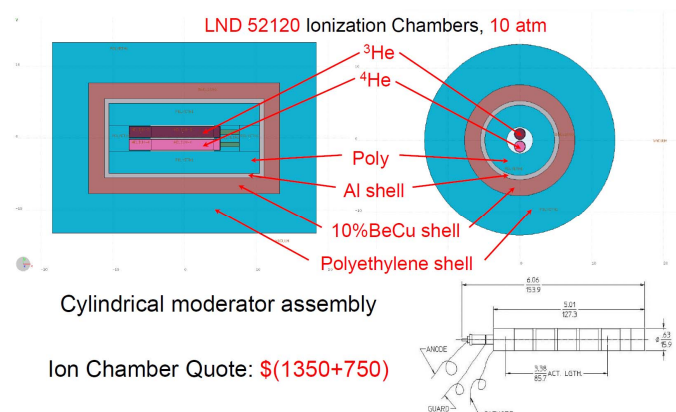


Fig. 1. Schematics of the NDX dose rate meter prototype. The ^3He and ^4He gas filled ionization chambers are placed in the center of the series of moderator shells, made consecutively of polyethylene, Aluminum, Be-Cu alloy, and polyethylene.

The prototype design schematic is shown in Fig. 1.

The NDX detector was modeled using FLUKA Monte Carlo simulation code [6,7], and the design parameters were optimized for constant response of the detector to the neutron

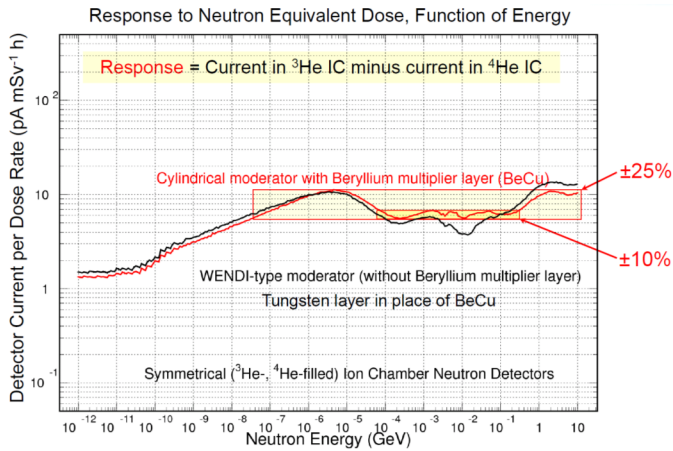


Fig. 2. FLUKA calculation of the NDX (red) and WENDI-type [3-5] (black) detector responses, function of neutron energy, measured as the difference of the ^3He and ^4He IC currents in pA per equivalent dose rate in mSv/h.

irradiation field in a wide energy range. Fig. 2 illustrates the calculations, and gives comparison of its response to the response of WENDI-type [3-5] neutron moderator design.

Main difference between the NDX and WENDI design is using the Be-Cu alloy as the intermediate layer in the NDX, as opposed to the Tungsten alloy layer used for WENDI. Use of Beryllium helps to increase and linearize the response around 10 MeV, and also makes the response closer to linear at high (GeV-range) energies.

III. PROTOTYPE TESTS AT JLAB

Two prototype detectors have been assembled, tested, and calibrated at JLab, exhibiting characteristics close to expectations in terms of sensitivity to neutrons and photons, and in terms of stability.

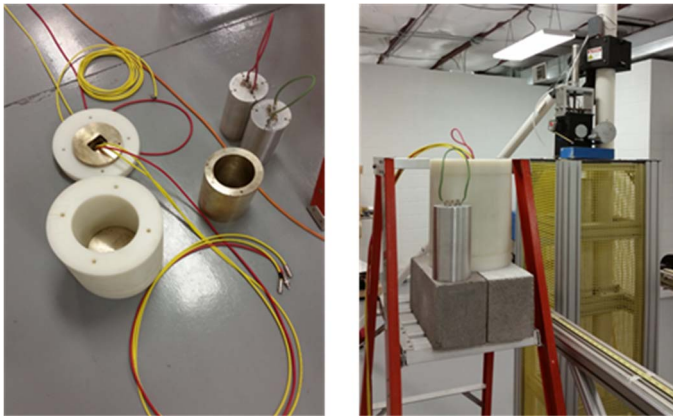


Fig. 3. NDX detector parts and assembly, shown at the JLab's Irradiation Facility before the detector calibration in the neutron field from the AmBe neutron source.

A sensitive electrometer-type current readout [8] is used, with a long-term stability better than 0.1 pA. The currents in ^3He and ^4He ion chambers agree within 10% in photon-only irradiation field. The calibration of the ^3He - filled ion chambers in the test neutron fields (see Fig. 3) resulted in the values of the calibration coefficients of about 0.11 mSv/h per pA.

The approximate calibration of the photon channel was performed at the photon irradiator installation at JLab, in the radiation field from the ^{137}Cs calibrated source, at approx. 1 Sv/h field. Distinct from the neutron channel with an almost uniform response of the dose readout to different neutron energies, the photon channel main purpose is to evaluate and subtract the contribution of photon background (current in ^4He IC) from the neutron signal (current in ^3He IC) in the mixed photon and neutron radiation fields. The photon response is calibrated at the ^{137}Cs 662 keV photon line, but should be corrected to the photon energy distribution if one needs to evaluate the absolute dose rate in an arbitrary photon field. Without such correction we refer to the photon dose rates measured by the NDX detectors, as “uncorrected”. In many practical situations when the shape of the photon spectra remains mostly unchanged (like in typical high energy electromagnetic cascades in thick targets), the photon channel will respond mostly linearly to the absolute real photon dose rates, and its calibration can be used for an approximate dose evaluation.

IV. LONG TERM OPERATIONS

After the period of initial testing in 2018 the detectors were installed in different locations at CEBAF to assess the stability of their long term operations and to learn about the various radiation conditions on site.

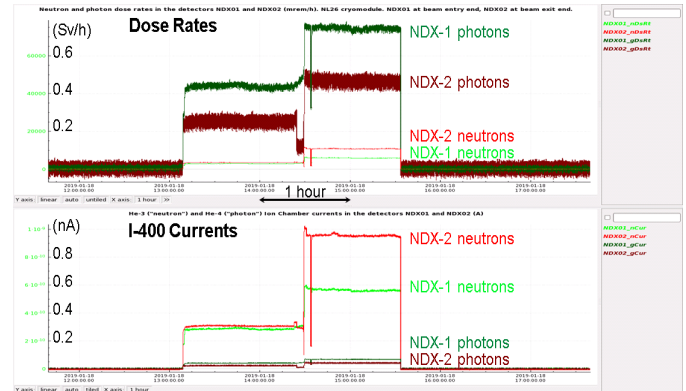


Fig. 4. The plot is an annotated computer screen snapshot of the monitoring and control system. NDX detectors' response during a couple hours of operations of one of the superconducting accelerating cryomodules at CEBAF North Linac is shown as a function of time. The bottom panel shows NDX IC currents, and the top panel illustrates the conversion of the currents into the measured radiation dose rates.

Fig. 4 shows the NDX detectors' response during several hours of operations of the superconducting accelerating cryomodules at CEBAF. The detectors were installed at the ends of one of the new C100 cryomodules in the accelerator tunnel. Upon turning on and ramping up the accelerating gradients, the electron field emission starts in the accelerating cavities, the electrons get accelerated, and may reach energies up to approx. 100 MeV, capable of producing neutrons and nuclear isotopes upon impact. The photon radiation field is much stronger than neutron radiation field in such conditions, because most of the total dose rate comes from the lower energy photons under the neutron production threshold. But the neutron dose rate is quite noticeable and is measured

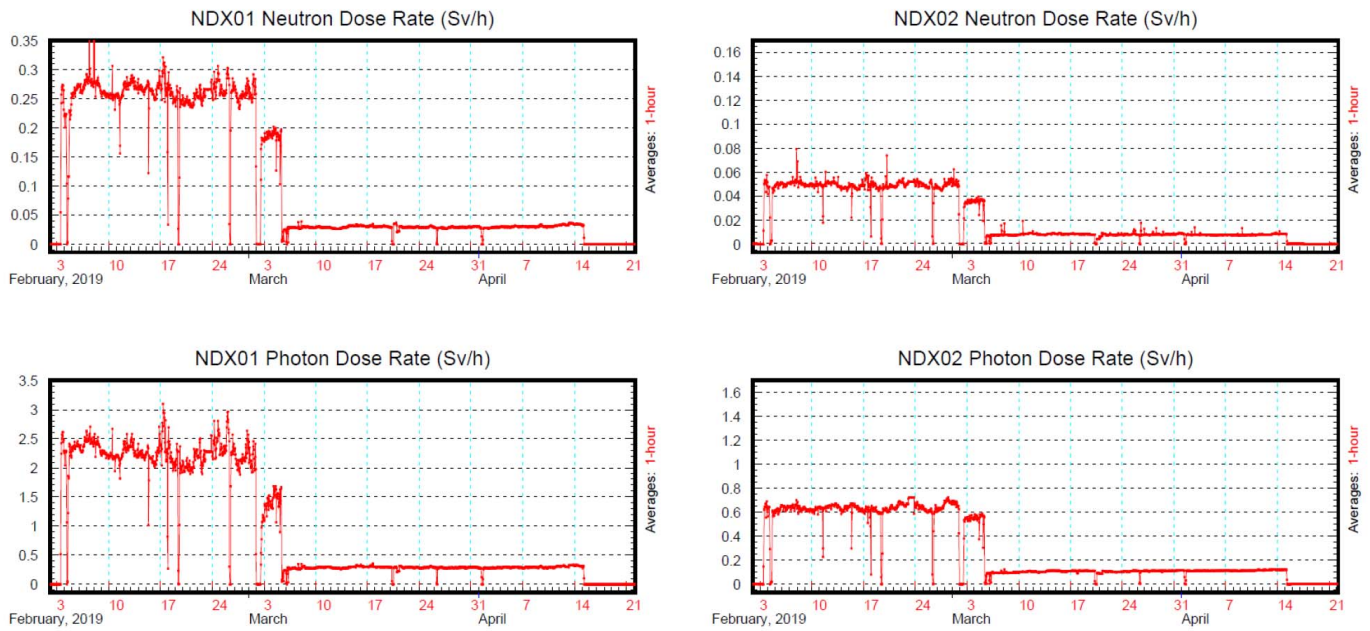


Fig. 5. Spring 2019 operations at CEBAF North Linac. Time histories of neutron and photon dose rate measurements (1-hour averages) using the two prototype NDX detectors placed next to one of the CEBAF accelerating cryomodules during operations in January-April 2019. Most of the average dose rate is generated by the relatively steady processes of electron field emission. Rare incidents of short accelerator beam losses were also registered, but they are practically not visible here due to averaging.

reliably. The photon detection in the NDX ICs is suppressed by the effective thick photon shielding around the ICs inside the detectors, and also by the relatively small ionization in the Helium gas. In the example shown (Fig. 4, bottom panel), the photon (^4He IC) currents are still smaller than the neutron (^3He IC) currents, indicating, however, that the real evaluated photon dose rates were much higher than neutron (see Fig. 4, top panel). In this example the correction to the neutron currents in ^3He from the photon background as measured by the ^4He currents was small, about 10%. That leads to the conclusion that the NDX detectors are capable of measuring neutron dose rates in conditions of overwhelming photon radiation fields (up to a factor of 100-500).

Fig. 5 illustrates the continuous three months of the NDX detectors' operations at the same tunnel location during the accelerator run period in the spring of 2019. During the first month of operations high accelerating gradients were used causing significant and not very stable levels of the field emission radiation around the location. After that period measures were taken to lower and optimize the operating gradients of the neighboring cryomodules, resulting in lowering the field emission radiation dose rates and stabilizing operations until the end of the run. During this deployment the NDX detectors also registered rare and short-duration beam loss events during which CEBAF's electron beam produced electromagnetic cascades in the beam line elements nearby, instantly increasing both photon and neutron production. Depending on the beam loss location, the measured dose rate increases could be extremely high, and the NDX detectors can serve to detect and characterize the loss. Particularly interesting application of the NDX neutron detectors would be their ability to localize events of high energy electron beam losses, using the feature of much more isotropic neutron

production compared to the pattern of the photon radiation field during such losses, which is very forward-directed. The string of NDX detectors placed along the beam line would be capable of locating the beam loss spot by balancing the observed neutron production in the neighboring detectors.

The test run has also demonstrated the long-term stability of the detectors which was found to be within 0.05 mSv/h, and much better short-term stability. The dynamic range of the detectors is expected to correspond to the large dynamic range generally accessible to ion chambers (5-6 orders of magnitude).

The detectors were also used for the characterization of radiation environments in the high power experimental halls at CEBAF during experiments capable of producing high radiation dose rates around the experimental targets and beam lines. So far such measurements at CEBAF were difficult. Many of the real-time neutron monitors available on the market are either not capable of operating in heavy photon radiation backgrounds, or fail due to radiation damage to the front end electronics. The NDX measurements make possible monitoring the stability, consistency, and the "health" of the experimental setups and running conditions, and provide alerts when running parameters are off limits.

Fig. 6 illustrates such monitoring during the run period in July-September 2019 in Hall A. The experiment used high power electron beams and relatively thick targets, and was designed to withstand the heavy radiation conditions. The monitoring of the radiation levels around the beam line helped to verify and benchmark the preliminary model assessments, and also to help identify conditions when intervention was needed. The NDX detectors were placed at two locations downstream the target Interaction Chamber, one (NDX1) closer to the target but heavily shielded, and the other (NDX2)

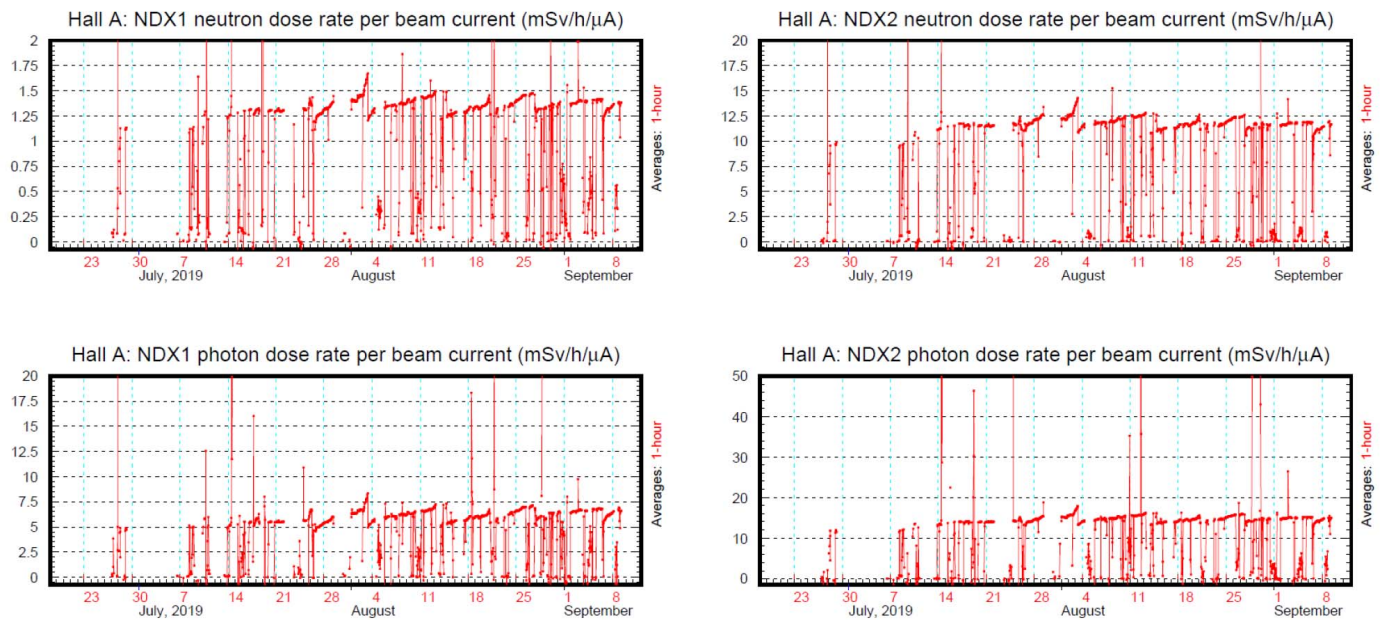


Fig. 6. July-September 2019 operations in the experimental Hall A at CEBAF. Shown are the time histories of neutron and photon dose rates (1-hour averages) relative to the beam current delivered to the hall, as measured by the two NDX detectors placed next to the beam line. 1 GeV electron beam currents varied, reaching up to 75 μA . Hours with no beam delivered are not included.

at the end of the beam line next to the entrance to the beam dump tunnel. NDX2 readings were generally much larger, but both detectors were sensitive to the beam and target parameters. The plot panels in Fig. 6 show ratios of the hourly average detectors' readings to the averaged beam current delivered to the hall. All hours in the period are shown in which beam current was delivered to the Hall, including various test and calibration setups. During the most of the beam time the experiment utilized relatively thick lead targets, producing significant radiation fields in the Hall. The beam currents were varying, reaching up to about 75 μA , but the ratios of the measured dose rates to the beam current were stable, however slowly drifting. The drifts were noticed to be correlated with slow deterioration of the targets in the beam, causing elevated scattered radiation in the hall, and the steps in the function correspond to the replacement of these targets. The dose rate measurements helped to characterize the state of the targets and optimize the operations. These measurements also serve as the data sets suitable for validation and proper

benchmarking of the evaluations and calculations of the radiation environment in the experimental Hall during the run. The detailed Monte Carlo simulation studies are under way.

Similar use of the NDX detector data is illustrated in Fig. 7. The detectors were installed in the experimental Hall C at CEBAF during the typical experiment with different targets. The NDX1 detector was placed at about 3 m from the target location, on the platform of the SHMS magnetic spectrometer. During the experiment SHMS was capable of rotating around the target center, to study electron scattering at different angles. A series of the detector readings normalized per beam current is presented in Fig. 7 as a function of calendar time, with the target and the spectrometer angle changes indicated in the plot. The target list included liquid deuterium, liquid hydrogen, and Aluminum, and the angle changed between 12 and 20 degrees, with the duration of different configurations about 1 hour each. One minute, ten minute, and hourly averages of the ratio are shown in the plot. Beam current was varying. The ratios demonstrate the stability of the detector readout, and the natural variations in the values, dependent on the type of the target, and the scattering angle. Neutron dose production on the deuterium target in forward direction is about twice the production on hydrogen, and Aluminum target showed lower rates because it was much thinner. The dose rates on hydrogen and deuterium are strongly dependent on the scattering angle, and neutron production from Aluminum practically does not depend on the angle in this range. Such behavior of the result is expected, and the data can be used for quantitative evaluation of the simulation models, in the planning of the future experiments.

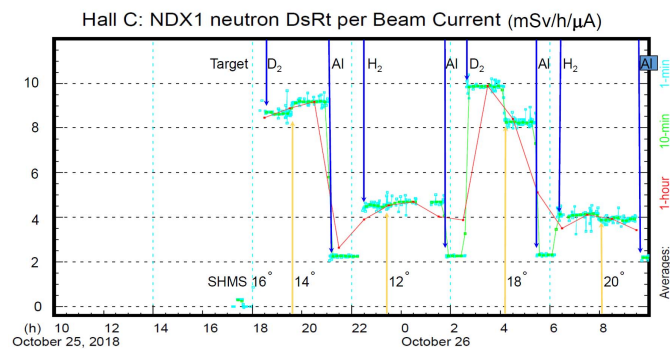


Fig. 7. The plot is similar to the top left panel of Fig. 6, but the measurements were performed in 2018 in the experimental Hall C at CEBAF, during the experiment on 10.6 GeV electron scattering from various targets. Beam currents were varying, reaching 60 μA . Arrows indicate changes of the targets and the angles at which the SHMS spectrometer was positioned.

V. CONCLUSION

Neutron dose rate meters based on the new NDX technology are suitable for neutron measurements in conditions of high radiation fields, including overwhelming photon radiation and pulsed beams. Uniform response function makes them applicable in high energy particle accelerators and irradiation facilities. The production version of the next generation of the detector is being developed at JLab, taking into account the lessons learned.

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